

Tesler's Law

What is it?

Complexity is inevitable in every system, no matter how much you simplify it.

Tesler's Law, also called the **Law of Conservation of Complexity**, says that every application has a certain amount of unavoidable complexity. You can't get rid of it—you can only shift where it lives: with the system, or with the user.

For example, shopping online for a jacket looks simple, but comparing sizes, styles, and prices reveals hidden layers of complexity beneath the clean interface.

History

Tesler's Law was introduced by **Larry Tesler**, a computer scientist who worked at **Xerox PARC**, **Apple**, and **Amazon**.

While developing early GUIs, Tesler noticed that removing visible complexity often just pushed it elsewhere—usually onto the user.

The Psychology Behind It

Humans crave simplicity, but true simplicity comes from managing complexity, not eliminating it. When designers oversimplify, they may remove important details or options that users need. The art lies in deciding **where** complexity should live—inside the system (through automation or logic) or with the user (through control and choice).

Why It Matters

- Simplifying one part often complicates another
 - Over-simplification can confuse users
 - Unclear structures increase frustration
 - Balance between ease of use and functionality builds trust
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How to Apply It

- **Minimize cognitive load** – Use chunking, progressive disclosure, and clear hierarchy.

- **Streamline workflows** – Remove unnecessary steps, provide shortcuts, and use visual cues.
 - **Design intelligent systems** – Use automation, personalization, or assistants like chatbots and guided tutorials.
 - **Provide clear instructions** – Tooltips, contextual help, and onboarding walkthroughs.
 - **Ensure smooth navigation** – Maintain clarity in menus and user pathways.
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Theory in Action

- **Netflix** simplifies discovery by automating recommendations using user data—keeping complexity behind the scenes.
 - **Tesla** hides technical processes from users by automating software updates, reducing their cognitive load.
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Final Thought

Complexity isn't the enemy—it's part of design.

The goal isn't to erase it but to decide *who* should handle it: the system or the user.

Well-balanced products feel simple because they make complexity invisible, not because it's gone.